
First-Divergence Model Diffing Reveals Upstream-Conditioned Late Readout in Base-to-Instruct Language Models

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Abstract

1 Late-layer patching is a natural way to localize how an instruction-following
2 checkpoint differs from its pretrained base, but a late patch can conflate two ques-
3 tions: what the late stack writes, and how much of that write is portable across
4 upstream states. We introduce **first-divergence model diffing**. At the first
5 shared-history prefix where a base/instruct pair prefers different next tokens, we
6 cross PT/IT upstream state with PT/IT late stack and score the IT-vs-PT token mar-
7 gin. The estimand measures a late-readout portability gap. Across five dense core
8 families, the IT-minus-PT late-stack effect is ‘+1.71’ logits (‘3.3x’) larger from
9 IT-shaped upstream state than from PT-shaped upstream state, with every family
10 positive. Targeted controls show the gap is not explained by broken hybrids, ar-
11bitrary token selection, or pre-late commitment. Depth and feature analyses give
12 a candidate-to-margin picture: middle MLP windows are relatively more identity-
13 selective, late/terminal windows are more margin-sensitive, and causally ranked
14 terminal crosscoder features partially mediate, inherit upstream conditioning, and
15 respond to preterminal state patches. The result is not explained by a fully portable
16 late-only instruction module; it is an upstream-conditioned late readout in released
17 dense base/instruct checkpoint contrasts.

18 1 Introduction

19 When an instruction-following checkpoint first chooses a different next token from its pretrained
20 base, where in the forward pass does that difference become a logit? Paired base/instruct check-
21 points make this question unusually clean: architecture and tokenizer are shared, but the released
22 descendants differ after instruction tuning, preference optimization, reinforcement-style training, or
23 a mixture of post-training stages. The object of study is therefore a **paired-checkpoint model diff**.

24 The tempting answer is “late layers.” Late transformer computation is close to the unembedding, and
25 prior work already makes late-stage refinement plausible: feed-forward layers promote vocabulary-
26 space concepts (Geva et al., 2022b), late layers sharpen or calibrate predictions (Lad et al., 2025;
27 Joshi et al., 2025), and instruction-tuned models show layer-structured task information (Zhao, Ziser,
28 and Cohen, 2024). But ordinary late-layer patching leaves a crucial ambiguity. A late stack can look
29 causal partly because it is paired with the upstream residual state it was trained to read, not because
30 its effect would transfer unchanged as a late-only update.

31 We test portability directly with **first-divergence model diffing**. For each prompt, we find the
32 earliest generated position where PT and IT prefer different next tokens under the same generated
33 history. Let those tokens be t_{PT} and t_{IT} . At that prefix, we cross upstream residual state (U_{PT}
34 or U_{IT}) with downstream late stack (L_{PT} or L_{IT}) and measure $\text{logit}(t_{IT}) - \text{logit}(t_{PT})$.

The key estimand is a difference-in-differences: how much larger is the IT-minus-PT late-stack replacement effect when the upstream state is IT-shaped rather than PT-shaped?

The answer is large, positive in every core family, and not a one-cell late-only effect. Across the five-family core set, the same IT late stack has a 3.3x larger margin effect from IT upstream than from PT upstream, producing a +1.71 logit interaction. The effect remains positive at generated positions ≥ 3 , under a common-PT readout, on a 32B family, under label-swap nulls, and under controls for pre-late token commitment, hybrid-state mismatch, and selected-token support.

This matters for model diffing practice. A late patch should not be interpreted as a fully portable late-layer mechanism unless the late effect is tested under both native and foreign upstream states.

The paper has three contributions.

1. **A paired-checkpoint first-divergence factorial.** We introduce a local counterfactual estimand at the exact token where PT and IT first disagree. It asks whether the IT-minus-PT late-stack replacement effect is portable across upstream states or amplified by IT-shaped upstream state.
2. **A validation ladder for the estimand.** We test the practical artifact explanations: broken hybrids, arbitrary selected-token support, pre-late commitment, readout choice, label orientation, position support, and family heterogeneity. The main text gives the decisive checks; the appendices give provenance.
3. **A depth and feature bridge.** Middle-positioned MLP substitutions are relatively more candidate/identity-selective, while late and terminal MLPs dominate margin/readout. Terminal crosscoders then connect the window-level result to sparse features that partially mediate the readout interaction, matter more under IT-shaped upstream state, partially rescue the weak hybrid, and respond to preterminal computation patches.

Scope is intentionally local. We use **IT** as shorthand for instruction-following post-trained descendants, but the recipes are heterogeneous; the empirical claim is about released dense base/instruct checkpoint contrasts, not one training algorithm. Causal language is readout-scoped: replacing an upstream state, late stack, or MLP component changes a specified next-token readout in a constructed forward pass. We do not claim a universal instruction-following circuit or full circuit reconstruction.

2 Setup

2.1 Model Sets and Statistical Reporting

The main factorial uses five dense PT/IT pairs: Llama 3.1 8B, Qwen 3 4B, Mistral 7B, OLMo 2 7B, and Qwen2.5 32B. We call this the **Core-5 set**. Several support analyses were run before the 32B extension and use the four smaller families: Llama, Qwen 3, Mistral, and OLMo. We call this the **Core-small support set**.

Prompt-bootstrap intervals quantify conditional precision on the sampled prompts and released checkpoints. They are not estimates over all possible model families, post-training recipes, or prompt distributions. For headline family-generalization claims, we therefore report the Core-5 mean and family-level range or median where it helps interpretation.

Table 1: Minimal reproducibility snapshot. The table below gives the fixed ingredients needed to audit the main factorial. Revision prefixes are unique abbreviations of the pinned Hugging Face revisions; Appendix A lists the full 40-character pins.

Family	Checkpoint pair (PT -> IT)	Prompt set	Prompts -> valid events	First-div. position 0 / ≥ 3 / ≥ 5	Late stack
Llama 3.1 8B	Llama-3.1-8B@d04e592 -> Instruct@0e9e39f	holdout-600	600 -> 600	59.3% / 28.0% / 17.3%	layers 19-31
Qwen 3 4B	Qwen3-4B-Base@906bfd4 -> Qwen3-4B@1cfa9a7	holdout-600	600 -> 600	48.7% / 35.2% / 23.5%	layers 22-35
Mistral 7B	Mistral-7B-v0.3@caa1feb -> Instruct@c170c70	holdout-600	600 -> 597	30.8% / 31.3% / 20.1%	layers 19-31
OLMo 2 7B	OLMo-2-1124-7B@7df9a82 -> Instruct@470b1fb	holdout-600	600 -> 586	60.1% / 27.1% / 16.0%	layers 19-31
Qwen2.5 32B	Qwen2.5-32B@1818d35 -> Instruct@5ede1c9	full-1400	1400 -> 1397	38.8% / 45.0% / 30.6%	layers 38-63

All rows use raw-shared prompts, greedy top-1 first-divergence search with a real-token mask, and at most 128 generated tokens. Bootstrap unit is the prompt cluster within family. Appendix A gives prompt mixes and full revision pins; Appendix B gives the main artifact roots.

2.2 First-Divergence Factorial

For each prompt, generate with PT and IT until the first shared-history prefix where their top-1 next tokens differ. Let those tokens be t_{PT} and t_{IT} , and define

$$Y(U, L) = \text{logit}(t_{IT}) - \text{logit}(t_{PT}).$$

Larger Y means the hybrid forward pass favors the IT divergent token. At a pre-specified late boundary, run the four cells below:

Upstream state	PT late stack L_{PT}	IT late stack L_{IT}
PT upstream U_{PT}	$Y(U_{PT}, L_{PT})$	$Y(U_{PT}, L_{IT})$
IT upstream U_{IT}	$Y(U_{IT}, L_{PT})$	$Y(U_{IT}, L_{IT})$

The primary estimand is:

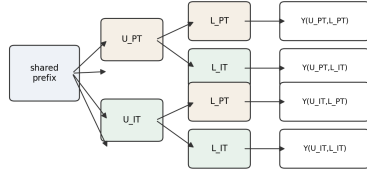
$$[Y(U_{IT}, L_{IT}) - Y(U_{IT}, L_{PT})] - [Y(U_{PT}, L_{IT}) - Y(U_{PT}, L_{PT})].$$

Equivalently: first measure the IT-minus-PT late-stack replacement effect under each upstream state; then compare those two effects. This subtracts the PT-late-stack baseline within each upstream row, so the result is not merely a native-stack-vs-foreign-stack comparison. Common-IT and common-PT readouts score all four cells with one fixed final norm, $1m_head$, and real-token mask. Unless stated otherwise, main factorial numbers use common-IT readout.

All raw-shared first-divergence and residual-state runs force both PT and IT branches to raw text and validate identical raw prompt token IDs before comparing residual states. Position 0 is therefore the first generated token after the full raw prompt, not a chat-template artifact.

A. Four hybrid passes at the first divergence

Same raw prompt and generated prefix. The readout is $Y = \text{logit}(t_{IT}) - \text{logit}(t_{PT})$.



$$\text{Interaction} = [Y(U_{IT}, L_{IT}) - Y(U_{IT}, L_{PT})] - [Y(U_{PT}, L_{IT}) - Y(U_{PT}, L_{PT})]$$

Interpretation: how much more the IT late stack helps when it reads an IT-shaped upstream state.

B. Concrete divergent-token examples

Model	Prompt fragment	pos.	PT token	IT token
Mistral 7B	Question: What is the difference between a...	4	'radio'	'digital'
OLMo 2 7B	Question: What causes the seasons? Answer:	3	'axis'	'tilt'
Owen 3 4B	Write a startup pitch for a time capsule...	5	'1'	'3'
Llama 3.1 8B	Question: My landlord is refusing to fix the...	13	'legal'	'several'

Examples are illustrative; all statistics use the full first-divergence support.

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Figure 1: First-divergence schematic and token examples. Panel A shows the four hybrid passes used to estimate the upstream x late interaction. Panel B gives illustrative divergent-token pairs; all quantitative claims use the full support in Table 1.

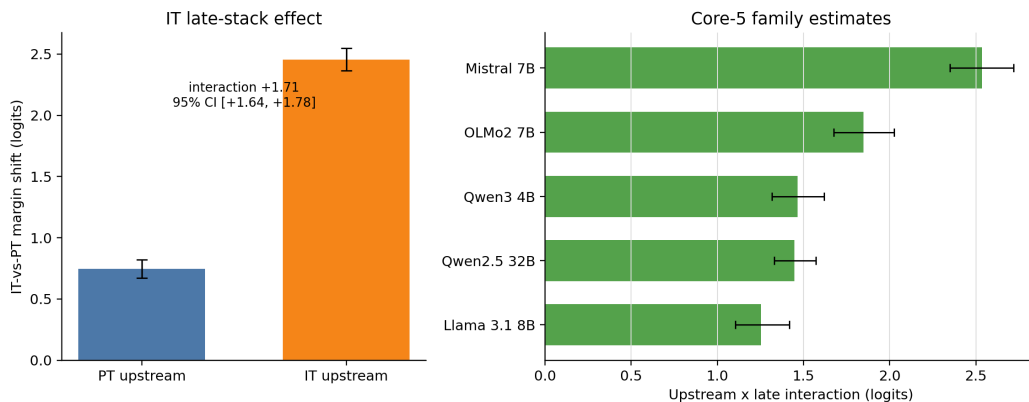
3 Results

3.1 Main First-Divergence Factorial

At the first natural PT/IT disagreement, the IT late stack is only partially portable. In the Core-5 set, replacing the PT late stack with the IT late stack shifts the IT-token margin by +0.75 logits from PT upstream, but by +2.46 logits from IT upstream. The +1.71 logit gap means the matched-context effect is 3.3x larger.

The sign pattern is consistent across families: every Core-5 interaction is positive, ranging from +1.25 to +2.53 logits. The amplification scale is the cleanest magnitude reference because it compares the same late-stack replacement under two upstream states; Appendix B reports native-shift shares and family-level ranges.

The interaction is also not confined to response openings: generated position ≥ 3 and ≥ 5 subsets remain positive, though smaller and thinner.



Scope/readout	Late effect from PT upstream	Late effect from IT upstream	Interaction	Amplification
Core-5, common-IT	+0.75 [+0.67, +0.82]	+2.46 [+2.36, +2.55]	+1.71 [+1.64, +1.78]	3.3x
Core-5, common-PT	+0.74 [+0.67, +0.81]	+2.48 [+2.38, +2.57]	+1.74 [+1.67, +1.81]	3.4x
Qwen2.5-32B only	+0.98 [+0.88, +1.08]	+2.42 [+2.26, +2.59]	+1.45 [+1.32, +1.57]	2.5x

On the local token-margin scale, +1.71 logits is substantial: it multiplies the odds of τ_{IT} over τ_{PT} by about 5.5x within the constructed contrast, and it is 34.6% of the native PT->IT diagonal margin shift on the same support. This is still a token-level mechanistic quantity, not a deployment-level behavior estimate. §3.2 adds a behavioral bridge showing that the same hybrid readout pattern changes immediate token choice and short-continuation preferences.

This is the paper's central claim: at natural base/instruct disagreement points, the stable object is the upstream x late interaction, not the portable component alone. On a factual/reasoning stress test, for example, the late-only effect from PT upstream is negative (-1.18), while the interaction remains positive (+1.81).

3.2 Validation Ladder

The first-divergence factorial intentionally selects a high-signal disagreement point and constructs hybrid forward passes. No patching experiment makes those hybrids literally natural trajectories. The validation question is narrower: do the main artifact explanations make the estimand uninformative? Four checks answer no; Appendix C gives the full audit trail.

Concern	Main evidence
The hybrid state is broken or generically off-manifold.	Native diagonals reconstruct exactly; PT->IT boundary interpolation is smooth and positive; signed permutations reconstruct exactly.
First divergence is an arbitrary selected token pair.	Random local disagreements from the same prompts retain only 56% of the first-divergence interaction; pre-divergence interaction remains positive.
The upstream state has already decided before the late stack.	The interaction remains positive when the IT boundary readout does not yet favor τ_{IT} and in the lowest boundary-ordered readout.
The logit margin does not affect generated behavior.	With the IT late stack fixed, IT-shaped upstream state raises τ_{IT} top-1 selection from 24.5% to 97.6%; pairwise judgments align.

Secondary checks agree: common-PT readout gives the same answer, later-position subsets remain positive, the label-swap null passes, and every core family has a positive interaction. Together, these controls make the practical artifact explanations unlikely. The exact logit magnitude remains intervention-scoped, and the short continuations are still constructed hybrid rollouts, but the upstream-conditioned late-readout pattern is robust under the tests that would otherwise explain it away.

3.3 Depth Anatomy: Candidate-to-Margin Handoff

The interaction has a consistent depth anatomy. Middle-positioned MLP substitutions transfer divergent-token identity more often than late substitutions, while late windows dominate margin/readout. Terminal-depth audits sharpen the same story: the last few blocks preserve a large readout subcomponent but transfer token identity poorly. The handoff is therefore operational rather

137 than a complete circuit: middle windows are relatively more candidate/identity-selective, while late
 138 and terminal windows are more margin/readout-sensitive.

Evidence

Middle vs late identity transfer

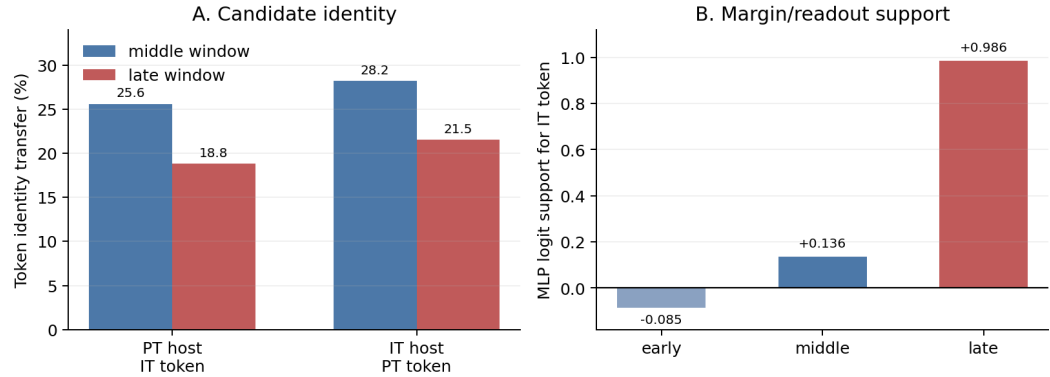
Native IT MLP margin support

PT-host late insertion

Terminal-depth audit

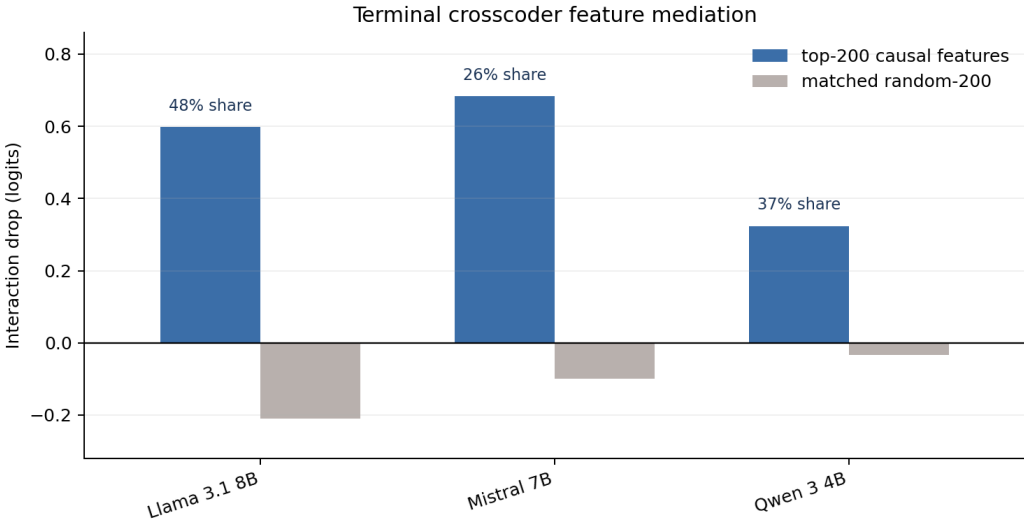
Terminal feature mediation, gating, rescue, and handoff top causal features account for 26–48%; causal gate beats matched random; direct feature rescue is +0.49; mid-to-preterm

Depth anatomy: identity before margin



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140 Terminal crosscoders make this handoff visible at feature level. We train paired PT/IT BatchTopK
 141 crosscoders with a shared latent dictionary and separate PT/IT decoder branches on terminal MLP
 142 outputs, rank features by held-out causal effect, and ablate their IT-branch decoder contribution
 143 inside the terminal IT stack. The fixed top-200 causal subset accounts for 26–48% of the terminal
 144 readout interaction, while matched-random sets have the wrong sign or near-zero effect. Appendix
 145 D gives reconstruction, sparsity, and training details.



146

147 The same features inherit the upstream conditioning seen at window level. Ablating the top-200
 148 causal terminal features hurts the U_IT,L_IT readout much more than the U_PT,L_IT readout,
 149 beating matched-random features in all three clean terminal-crosscoder families (family means
 150 +1.25, +0.98, +0.43 logits). Conversely, patching their native U_IT,L_IT activations into the
 151 weak U_PT,L_IT hybrid rescues +0.49 logits and beats both matched-random and same-delta ran-
 152 dom rescue. This rescue is deliberately harder than ablation and is not full reconstruction, but it

153 shows that the upstream-conditioned terminal features recover a measurable slice of the missing
154 IT-token margin.

155 Finally, a direct handoff test perturbs upstream computation and re-measures the same terminal
156 features. Injecting IT mid-to-preterminal computation into the weak U_PT,L_IT hybrid rescues
157 +1.71 logits of IT-token margin, with +0.13 logits mediated by the selected terminal features. The
158 reverse intervention, replacing the same preterminal computation in native IT with PT computation,
159 causes a +3.57 logit drop, with +0.53 mediated. A terminal-entry patch gives the expected upper
160 bound (+5.15 total, +0.71 mediated). Thus the feature bridge is not only terminal: preterminal
161 state changes drive a measurable part of the terminal sparse-feature readout.

162 Held-out autointerp makes the mediated feature set readable but not load-bearing. Across 225 in-
163 terpreted features from the clean terminal-crosscoder families, validation reaches mean AUROC
164 0.886; we use these labels descriptively while the causal evidence comes from feature edits. As a
165 targeted semantic check, a predeclared `structure_readout` bucket gives a monotone positive edit
166 across clean crosscoder families; Appendix D reports the full dose response and controls.

167 3.4 What the Factorial Separates

168 The first-divergence event is a distributional fact: the released PT and IT checkpoints prefer different
169 next tokens at that prefix. The factorial asks a different question: once the token contrast is fixed,
170 how much of the IT late-stack effect ports across upstream states? The answer is: some, but not
171 most. The IT late stack can add IT-token margin from PT upstream, but most of its matched-context
172 readout effect appears only when the upstream state is IT-shaped.

173 This distinction is why the result is more than "PT and IT differ." Random local disagreements, pre-
174 divergence future-token scoring, and pre-late commitment controls all reduce or preserve the interac-
175 tion in the predicted directions. The feature bridge then shows that part of the window-level interac-
176 tion is carried by terminal sparse features that are more causally important under IT-shaped upstream
177 state, can partially rescue the weak PT-upstream hybrid, and respond when upstream/preterminal
178 computation is patched.

179 Two supporting analyses stay in the appendix. Endpoint-matched late-refinement signatures explain
180 why late/terminal windows were a good target: after matching on final entropy, confidence, and
181 margin, IT late predictions remain farther from their own final readout than PT late predictions under
182 both raw and tuned probes (+0.425 and +0.762 nats; Appendix E). This is a localization signature,
183 not the mechanism itself. An OLMo-2 Base/SFT/DPO/RLVR lineage shows that the same estimand
184 can be tracked along one released post-training path, without turning that single path into a general
185 stage attribution.

186 Contemporaneous token-level RLVR work reaches a complementary conclusion from the outside:
187 Sparse but Critical shows that a small set of shifted token decisions can carry large downstream
188 effects under cross-sampling interventions (Meng et al., 2026). Our question is internal and paired-
189 checkpoint: at the selected token where behavior first changes, how does the IT-token preference
190 become a logit inside the model?

191 4 Related Work

192 **Late refinement and FFN readout.** Feed-forward layers promote vocabulary-space concepts and
193 progressively refine predictions (Geva et al., 2022a,b). Layerwise intervention studies describe
194 late residual sharpening (Lad et al., 2025), calibration analyses find an upper-layer confidence-
195 adjustment phase (Joshi et al., 2025), and tuned lenses operationalize layerwise prediction refine-
196 ment (nostalgebraist, 2020; Belrose et al., 2023). These works establish late refinement as plau-
197 sible. Our contribution is to measure how it differs across paired PT/IT checkpoints at natural
198 first-divergence tokens.

199 **Post-training model diffs.** Wu et al. (2024) study behavioral shifts from language modeling to
200 instruction following. Du et al. (2025) compare base and post-trained checkpoints mechanistically
201 across knowledge, truthfulness, refusal, and confidence. Zhao, Ziser, and Cohen (2024) show layer-
202 structured task information in instruction-tuned models. Sparse but Critical analyzes RLVR as sparse
203 token-level distribution shifts whose cross-sampled substitutions affect reasoning trajectories (Meng

et al., 2026). We add an internal paired-checkpoint counterfactual: at the first token where PT and IT disagree, how much of the IT-token margin comes from a portable late-stack effect, and how much requires IT-shaped upstream state?

Activation patching and feature-level model diffing. Activation patching requires care because metric choice, intervention direction, and off-manifold hybrids affect interpretation (Heimersheim and Nanda, 2024). We therefore report intervention-scoped readout effects and validate them with readout swaps, diagonal reconstruction, interpolation, low-anomaly filtering, label-swap nulls, signed-permutation controls, and random-disagreement baselines. Cross-model activation patching across base and fine-tuned variants is the closest methodological precedent (Prakash et al., 2024): their target is an entity-tracking mechanism and whether fine-tuning enhances it, while ours is the first natural PT/IT next-token disagreement and how much the late readout margin ports across upstream states. Sparse crosscoders provide a complementary route for model diffing (Lindsey et al., 2024), with known sparsity-artifact pitfalls (Minder et al., 2025). We use crosscoders after the factorial rather than as a standalone diff: causally ranked terminal features partially mediate the interaction, matter more under IT-shaped upstream state, and partially rescue the weak PT-upstream hybrid.

Automated feature interpretation. LLM-based neuron interpretation commonly follows a generate-and-score pattern: propose a natural-language feature hypothesis, then test it on held-out examples (Bills et al., 2023; Huang et al., 2023). We follow that pattern for terminal crosscoder features. Labels are descriptive evidence; feature ablations provide the causal evidence.

Novelty. Several ingredients have precedents: late refinement, FFN vocabulary promotion, activation patching, global base/instruct diffing, and sparse model-diff features. The new object is the paired-checkpoint first-divergence factorial estimand. It measures the portability gap in the IT-minus-PT late-stack replacement effect at the natural token where PT and IT first disagree. The feature-level contribution is the bridge from this window-level interaction to terminal sparse features: the same causally ranked features mediate part of the readout interaction, are selectively necessary under IT-shaped upstream state, partially rescue the missing margin when inserted into the PT-upstream hybrid, and are partly driven by upstream/preterminal computation.

5 Discussion, Scope, and Next Tests

5.1 Interpretation

The interpretation suggested by these results is that released base-to-instruct transitions reshape how existing computation is read out, rather than adding a portable late instruction module. This is compatible with a broader view of post-training as selecting, amplifying, or consolidating reachable behaviors from the pretrained model, but our evidence is narrower: it concerns dense base/instruct checkpoint pairs at first-divergence token readouts. RLVR analyses such as Sparse but Critical are complementary context because they study token-level post-training shifts from outside the model; our factorial asks where the selected token preference becomes a logit inside paired checkpoints.

The OLMo-2 lineage illustrates how the same estimand can be reused along one released post-training path. On fixed Base->RLVR first-divergence support, the measured interaction is already partly present after SFT and close to final after DPO, with the final RLVR/Instruct checkpoint strongest. This is a case study in applying the diagnostic across stages, not a general attribution of which post-training stage causes the effect.

5.2 Scope

The scope is narrow by design. First, the estimand is local to first-divergence next-token readouts: it targets the earliest point where a released PT/IT pair changes preference under shared history, not an average over all model behavior. Position-stratified and domain stress tests show that the interaction persists beyond the earliest generated positions, while its magnitude varies with prompt domain and generated position.

Second, the interventions are window-level compatibility tests. Hybrid-state validation makes practical artifact explanations unlikely, and terminal crosscoders provide concentrated feature-level mediation, rescue, and upstream-handoff checks, but we do not recover a full circuit or prove an on-

manifold natural-trajectory effect size. The headline logit margins are dimensionalized as odds and native-shift shares in §3.1; they should not be read as direct estimates of completion-level behavior.

Third, the empirical scope is five dense core PT/IT pairs. DeepSeek-V2-Lite stays appendix-only because MoE routing and expert swaps require different controls. Architecture and MoE generalization are therefore next-step questions, not claims made by the core dense-family result. The OLMo-2 result is a fixed-support case study of one released lineage, not a universal stage attribution.

5.3 Practical Implications and Next Tests

The immediate practical implication is methodological: late-patching, steering, and model surgery studies on paired checkpoints should test whether late effects are portable across upstream states before treating them as standalone late-layer mechanisms.

The most direct next test is to make the upstream side sparse as well: train or interpret middle/preterminal features directly and test which of them drive the upstream-conditioned terminal features.

6 Conclusion

First-divergence model diffing turns a vague question -- "do late layers explain the base-to-instruct difference?" -- into a paired-checkpoint counterfactual. At the first token where released PT and IT checkpoints disagree, the IT-minus-PT late-stack replacement effect is much larger from IT-shaped upstream state than from PT-shaped upstream state. The result is positive across the five dense core families and robust to targeted checks for readout choice, selected-token support, pre-late commitment, label orientation, and hybrid-state failure. The depth anatomy is graded: middle windows are relatively more identity-selective, while late and terminal windows are more margin/readout-sensitive. Terminal sparse features carry a concentrated bridge: they are more causally important under IT-shaped upstream state, their native-IT activation pattern partially rescues the weak PT-upstream hybrid, and upstream/preterminal patches drive a measurable part of their mediated effect. The resulting picture is not explained by a fully portable late-only update; it is an upstream-conditioned late readout pattern in released dense base/instruct checkpoint contrasts.

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332 *2024*.
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334 Happens in Instruction-Tuned Large Language Models. *EMNLP 2024*.

Appendix Roadmap

The main text is written around stable claim names. For details, start with Appendix B for the main four-cell factorial, Appendix C for validation controls, and Appendix D for depth/feature anatomy. Numeric run IDs appear only in file paths or script names; they are provenance labels, not concepts the reader needs to parse.

Claim	Main location	Appendix	Primary artifacts/scripts
Minimal reproducibility snapshot	§2.1	A, B, H	model registry, dataset manifests, raw first-d
Core-5 first-divergence interaction and amplification scale	§3.1	B	results/paper_synthesis/exp23_cor
Validation ladder	§3.2	C	hybrid-state validation, random-disagreement
Behavioral bridge from margin to output	§3.2	C	one-step top-1/token-rank audit and short-hy
Depth and terminal anatomy	§3.3	D	identity/margin handoff, terminal-depth audi
Terminal feature mediation, upstream-conditioning, rescue, handoff, and structure-bucket validation	§3.3	D	terminal crosscoder synthesis, hardening run
Late refinement/readout signatures	§3.4	E	endpoint-matched KL, late MLP random co
OLMo staged case study	§3.4	F	fixed-support Base/SFT/DPO/RLVR stage d
Architecture and MoE scope	§5	G	dense/MoE scope note

Prompt-bootstrap CIs in the main text are conditional precision estimates over sampled prompts and released checkpoints. They are paired with family-level summaries or family ranges where a claim could otherwise be mistaken for a population-level model-family generalization.

Appendix A: Model Scope and Statistical Reporting

Core-5 set. Llama 3.1 8B, Qwen 3 4B, Mistral 7B, OLMo 2 7B, and Qwen2.5 32B. Qwen2.5 32B is included as the scale check in the core first-divergence synthesis.

Core-small support set. Llama 3.1 8B, Qwen 3 4B, Mistral 7B, and OLMo 2 7B. Supporting identity/margin, terminal MLP, crosscoder, and KL analyses use this smaller-family scope unless explicitly marked otherwise. The Qwen2.5 32B pair is included in the main factorial and omitted from these support analyses for compute.

Prompt sets. The holdout-600 prompt mix is GOV-CONV/GOV-FORMAT/SAFETY = 300/150/150. The full-1400 mix adds factual, reasoning, register, and baseline-easy prompts: CONTENT-FACT/CONTENT-REASON/GOV-FORMAT/GOV-CONV/SAFETY/GOV-REGISTER/BASELINE-EASY = 300/200/250/300/150/100/100.

Family	PT checkpoint	PT revision	IT checkpoint	IT revision
Llama 3.1 8B	meta-llama/Llama-3.1-8B	d04e592bb4f6aa9cfee91e2e20afa771667e1d4b	meta-llama/Llama-3.1-8B-Instruct	0e9e39f249a169
Qwen 3 4B	Qwen/Qwen3-4B-Base	906bfd4b4dc7f14ee4320094d8b41684abff8539	Qwen/Qwen3-4B	1cfa9a72089121
Mistral 7B	mistralai/Mistral-7B-v0.3	caa1feb0e54d415e2df31207e5f4e273e33509b1	mistralai/Mistral-7B-Instruct-v0.3	c170c708c41dac
OLMo 2 7B	allenai/OLMo-2-1124-7B	7df9a82518afdecae4e8c026b27adccc8c1f0032	allenai/OLMo-2-1124-7B-Instruct	470b1fba1ae015
Qwen2.5 32B	Qwen/Qwen2.5-32B	1818d35814b8319459f4bd55ed1ac8709630f003	Qwen/Qwen2.5-32B-Instruct	5ede1c97bbab6c

The Core-5 synthesis combines stored per-family prompt-bootstrap estimates for the five families in §2.1. The amplification ratio reported in §3.1 is:

$$\text{late-stack amplification} = [Y(U_{IT}, L_{IT}) - Y(U_{IT}, L_{PT})] / [Y(U_{PT}, L_{IT}) - Y(U_{PT}, L_{PT})].$$

This is the main scale reference because it compares the same IT late-stack replacement under the two upstream states. We also report a secondary native-shift scale, computed inside the same 2x2:

$$\text{native PT} \rightarrow \text{IT diagonal margin shift} = Y(U_{IT}, L_{IT}) - Y(U_{PT}, L_{PT}) = \text{late_weight_effect} + \text{upstream_context_effect}.$$

The reported interaction share is $\text{interaction} / \text{native diagonal margin shift}$. This is a scale reference, not a claim that the interaction linearly decomposes all behavioral difference.

Appendix B: Main First-Divergence Factorial Audit Trail

Primary Core-5 artifacts:

- results/paper_synthesis/exp23_core5/exp23_core5_core_effects.csv
- results/paper_synthesis/exp23_core5/exp23_core5_family_effects.csv
- results/paper_synthesis/exp23_core5/exp23_core5_position_sensitivity.csv
- results/paper_synthesis/exp23_core5/exp23_core5_interaction.png

Raw first-divergence record mirrors:

- Core-small support: gs://pt-vs-it-results/results/exp23_midlate_interaction_suite/exp23_dense5_full
- Qwen2.5 32B support: gs://pt-vs-it-results/results/exp24_32b_external_validity/exp24_qwen25_32b_full

Main Core-5 effects:

Scope/readout	PT-up late effect	IT-up late effect	Interaction	Amplification	Native shift	Share
Core-5, common-IT	+0.747 [+0.673, +0.821]	+2.456 [+2.364, +2.547]	+1.709 [+1.637, +1.780]	3.3x	+4.942	34.6%
Core-5, common-PT	+0.740 [+0.673, +0.807]	+2.477 [+2.384, +2.570]	+1.737 [+1.667, +1.807]	3.4x	+4.996	34.8%

Core-5 family-level common-IT interactions:

Family	Interaction	Native diagonal shift	Interaction share
Llama 3.1 8B	+1.253	+5.358	23.4%
Qwen2.5 32B	+1.446	+3.751	38.6%
Qwen 3 4B	+1.464	+3.938	37.2%
OLMo 2 7B	+1.847	+5.227	35.3%
Mistral 7B	+2.534	+6.437	39.4%

The Core-5 family interaction range is +1.253 to +2.534 logits, with median +1.464. Interaction share ranges from 23.4% to 39.4%.

The Core-small label-swap null is computed from:

- results/paper_synthesis/exp23_core_small_compatibility_permutation/

The content/reasoning stress test is:

- results/exp23_midlate_interaction_suite/exp23_content_reasoning_residual_20260427_0930_h100x8/a

On that support, the late-only PT-upstream term is -1.176 and the upstream x late interaction is +1.812 ([+1.721, +1.901]).

Qwen2.5 32B artifacts:

- results/exp24_32b_external_validity/exp24_qwen25_32b_full_eval_v21_20260427_194839/analysis/
- results/paper_synthesis/exp24_32b_external_validity/

Appendix C: Validation Controls

Hybrid-state validation. Primary artifacts:

- results/exp36_offmanifold_validation/exp36_offmanifold_dense5_full_a100x8_20260502_233904/analysis/
- results/exp36_offmanifold_validation/exp36_offmanifold_dense5_full_a100x8_20260502_233904/analysis/
- results/exp36_offmanifold_validation/exp36_offmanifold_dense5_full_a100x8_20260502_233904/analysis/
- results/exp36_offmanifold_validation/exp36_offmanifold_dense5_full_a100x8_20260502_233904/analysis/

Key checks, summarized over the Core-small support families:

Check	Result
Endpoint interaction, common-IT	+1.794
PT-to-IT interpolation slope	+1.831
Signed-permutation random/observed ratio	0.31x

Random-disagreement and pre-divergence controls. Primary artifacts:

- results/exp37_random_prefix_baseline/exp37_full_dense5_auth_xetfast_h100x8_20260503_002609/analysis/

395 • results/exp37_random_prefix_baseline/exp37_full_dense5_auth_xetfast_h100x8_20260503_002609/anal
396 • results/exp37_random_prefix_baseline/exp37_full_dense5_auth_xetfast_h100x8_20260503_002609/anal

Condition	Interaction	Share of first divergence
True first divergence	+1.794	100%
Random local disagreement, source-balanced	+1.005	56%
Random PT-rollout disagreement	+0.836	47%
Random IT-rollout disagreement	+1.190	66%
Pre-divergence prefix, future token pair	+0.059	3%

397 **Token-support control.** The no-GPU support control compares true first divergence to random
398 local PT/IT disagreements from the same prompts and generated rollouts.

399 • scripts/analysis/analyze_exp37_token_support_control.py
400 • results/exp37_random_prefix_baseline/exp37_full_dense5_auth_xetfast_h100x8_20260503_002609/anal

401 Random local disagreements are later and more content-token-heavy than first divergences, yet their
402 factorial interaction is smaller. This rules out the simple explanation that the first-divergence inter-
403 action is large only because of early-position or format-heavy support.

404 **Selected-support audit.** Primary artifacts:

405 • scripts/analysis/analyze_first_divergence_token_support.py
406 • results/first_divergence_token_support/dense5_llm_gpt55_20260503_121500/summary.json
407 • results/first_divergence_token_support/dense5_llm_gpt55_20260503_121500/token_support_report.md

408 The audit supports the main-text scope statement: first divergence is a targeted high-signal disagree-
409 ment support, and most judged events fall in substantive instruction, safety, formatting, response-
410 shaping, or semantic-content categories rather than pure surface formatting. We keep the full cate-
411 gory table in the artifact report rather than in the manuscript because it is a support audit, not part of
412 the evidence spine.

413 **Pre-late commitment control.** Primary artifacts:

414 • results/exp40_prelate_commitment_control/exp40_exp20_layerwise_proxy_20260503_110001/analysis/s
415 • results/exp40_prelate_commitment_control/exp40_exp20_layerwise_proxy_20260503_110001/analysis/e
416 • results/exp40_prelate_commitment_control/exp40_exp20_layerwise_proxy_20260503_110001/analysis/e
417 • results/exp40_prelate_commitment_control/exp40_exp20_layerwise_proxy_20260503_110001/analysis/p

418 The support-run control restricts to events where the IT boundary readout does not yet favor t_{IT} ,
419 bins events by IT boundary margin, and fits boundary-margin controls. In all three views the inter-
420 action remains positive. This rules out the simplest "the late stack is irrelevant because the boundary
421 already committed" reading without promoting these support-run magnitudes to Core-5 headline
422 estimates.

423 **Behavioral bridge.** This support analysis asks whether the logit-margin interaction maps onto
424 generated-token behavior. It uses the Core-small support families and the same first-divergence
425 prefixes as the main factorial.

426 • results/exp45_behavioral_bridge/exp45_full_a100x8_20260504_0652/analysis/report.md
427 • results/exp45_behavioral_bridge/exp45_full_a100x8_20260504_0652/analysis/one_step_effects.csv
428 • results/exp45_behavioral_bridge/exp45_full_a100x8_20260504_0652/analysis/behavioral_effects.csv
429 • results/exp45_behavioral_bridge/exp45_full_a100x8_20260504_0652/analysis/llm_judge_summary.json

430 One-step readout gives the cleanest bridge. With the IT late stack fixed, changing the upstream state
431 from PT-shaped to IT-shaped raises t_{IT} top-1 selection from 24.5% to 97.6%; the corresponding
432 pairwise logit(t_{IT}) > logit(t_{PT}) rate rises from 31.2% to 98.7%.

Metric	U_PT,L_IT	U_IT,L_IT	Gap
t_{IT} top-1 rate	24.5%	97.6%	+73.1%
t_{IT} top-5 rate	70.9%	100.0%	+29.1%
$t_{IT} > t_{PT}$ pairwise rate	31.2%	98.7%	+67.5%

433 Short hybrid rollouts give a complementary completion-level check. A blinded pairwise judge
434 (gpt-4.1-mini) prefers U_IT,L_IT over U_PT,L_IT as more instruction-following / assistant-

like in 71.0% of cases when ties/unclear are counted as 0.5 ([69.4%, 72.6%]). The judge also finds a smaller positive late-stack effect under both upstream states: 58.5% for U_PT,L_IT over U_PT,L_PT, and 57.8% for U_IT,L_IT over U_IT,L_PT.

We use this as a behavioral bridge only. The one-step result is closest to the factorial estimand; the judged continuations are constructed hybrid rollouts, not natural deployment trajectories.

Appendix D: Depth, Terminal, and Crosscoder Details

Identity/margin handoff artifacts:

- results/paper_synthesis/exp20_exp21_handoff_table.csv
- results/paper_synthesis/exp20_exp21_handoff_core_small.png
- results/exp20_divergence_token_counterfactual/factorial_validation_holdout_fast_20260425_2009_w
- results/exp21_productive_opposition/exp21_full_productive_opposition_clean_20260426_053736/anal
- results/exp21_productive_opposition/exp21_full_productive_opposition_clean_20260426_053736/anal

Main depth-anatomy quantities:

Readout	Early	Middle	Late / terminal	Interpretation
PT host: IT-token identity transfer	-	25.6%	18.8%	Middle substitutions transfer candidate identity more often.
IT host: PT-token identity transfer	-	28.2%	21.5%	Mirror direction gives the same identity pattern.
Pure IT MLP support for t_{IT}	-0.085	+0.136	+0.986	Native IT-token support is late-concentrated.
PT-host late MLP margin gain	-	-	+0.004	Late MLP updates alone are near zero in PT upstream state.
Source decomposition interaction	-	-	+0.360	MLP-level readout also shows context gating.

Terminal-depth and terminal-MLP artifacts:

- results/exp31_terminal_depth_factorial/exp31_terminal_depth_full_a100x4_localdisk_fixedsched_20
- results/exp31_terminal_depth_factorial/exp31_terminal_depth_full_a100x4_localdisk_fixedsched_20
- results/exp32_terminal_mlp_writeout/exp32_terminal_mlp_full_dense5_a100x8_w2_20260502_043950/an
- results/exp33_terminal_identity_margin/exp33_terminal_identity_margin_full_dense5_a100x8_overla

In the Core-small support set, the final-three stack retains 52% of the same-prompt full-late interaction; the final block alone retains 23%. Final-three MLP substitutions transfer IT-token identity 8.8% of the time, with terminal MLP margin interaction +0.524 ([+0.491, +0.558]). The final layer alone gives terminal MLP margin interaction +0.146 ([+0.128, +0.165]).

Terminal crosscoder mediation. Artifacts:

- results/paper_synthesis/exp34_dense5_final_readout_crosscoder/combined_dense5_20260503_0018/exp
- results/paper_synthesis/exp34_core_feature_mediation/terminal_crosscoder_core3_mediation.png
- results/exp38_qwen_olmo_final_layer_crosscoder_hardening/exp38_qwen_olmo_final_summary_20260503
- results/exp30_final_readout_crosscoder_mediation/exp30_l31_paperfaithful_runpod_20260502_012105

Family	Terminal scope	VE / density status	Top causal feature drop	Share of interaction	Matched random	Paper role
Llama	final 3 layers	VE min 0.774, LO 64, alive max 0.096	+0.599 [+0.469, +0.733]	48%	-0.209 [-0.255, -0.165]	clean
Mistral	final 3 layers	VE min 0.786, LO 64, alive max 0.089	+0.684 [+0.600, +0.764]	26%	-0.100 [-0.159, -0.044]	clean
Qwen	final 2 layers	layer VE 0.957/0.960 and 0.967/0.970	+0.324	37%	-0.033	clean

OLMo terminal crosscoder quality did not pass the predeclared reconstruction gate, so OLMo is excluded from feature-level claims.

The mediation curves sweep the number of ablated causally ranked features. The main table reports top-200 because it is fixed across families and far from the full-dictionary reconstruction setting; the curves show saturation rather than a single hand-picked feature count. The top-200 set is a small causally ranked subset of the crosscoder dictionary, not a full reconstruction. Recovering 26–48% of the exposed terminal interaction from this subset is therefore a concentration result; matched-random features do not reproduce it.

Family	top-200 share	top-500 share	Saturation read
Llama	48%	52%	modest additional distributed mass
Mistral	26%	29%	modest additional distributed mass
Qwen	37%	38%	mostly saturated by top-200

Terminal feature upstream-conditioning. Artifacts:

- results/exp42_terminal_feature_upstream_conditioning/exp42_full_4fam_h100x8_20260503_155212/ana
- results/exp42_terminal_feature_upstream_conditioning/exp42_full_4fam_h100x8_20260503_155212/ana
- results/exp42_terminal_feature_upstream_conditioning/exp42_full_4fam_h100x8_20260503_155212/ana
- results/exp42_terminal_feature_upstream_conditioning/exp42_full_4fam_h100x8_20260503_155212/ana

For the same causally ranked terminal features, we compare ablation effects in the U_IT,L_IT and U_PT,L_IT cells. The primary feature causal gate is:

[drop when ablating features in U_IT,L_IT] - [drop when ablating features in U_PT,L_IT].

Positive values mean the feature set matters more when the IT terminal stack receives IT-shaped upstream state. At top-200 features:

Metric	Estimate
Absolute causal feature gate, clean-family mean	+0.703
Causal gate minus matched-random features	+0.887
Causal gate minus top-active noncausal features	+1.495
Margin-weighted activation gate minus matched-random features	+0.520

Per-family gates:

Family	Absolute causal gate	Causal minus matched random
Llama	+0.922	+1.254
Mistral	+0.816	+0.982
Qwen	+0.370	+0.426

Raw decoder-weighted activation mass is not uniformly higher under IT-shaped upstream state across families. We therefore use the finite-difference causal gate as the primary upstream-conditioning result and the signed margin-weighted activation gate as supporting evidence.

Terminal feature rescue. This analysis tests a partial-sufficiency version of the same feature-level story. It runs on the three clean rescue families with quality-gated terminal crosscoders (Llama, Mistral, Qwen). OLMo is excluded because its current terminal crosscoder does not pass the reconstruction-quality gate needed for faithful feature-space rescue edits.

The edit takes the top-200 causal terminal feature activations from the native U_IT,L_IT pass and patches them into the U_PT,L_IT hybrid, decoded through the IT branch of the paired PT/IT crosscoder. The metric is rescued IT-token margin:

$Y(U_{PT,L_IT} + \text{rescued features}) - Y(U_{PT,L_IT})$.

Rescue metric, Llama/Mistral/Qwen family-balanced	Estimate	95% CI
Direct top-200 causal feature rescue	+0.494	[+0.451, +0.539]
Direct rescue fraction	8.1%	[5.5%, 10.3%]
Causal minus matched-random rescue	+0.561	[+0.510, +0.613]
Causal minus matched-random rescue fraction	10.8%	[7.6%, 13.7%]
Causal minus same-delta-random rescue	+0.471	[+0.427, +0.517]
Causal minus same-delta-random rescue fraction	8.3%	[5.7%, 10.6%]

Per-family direct rescue is Llama +0.627, Mistral +0.755, and Qwen +0.101 logits. The alpha-zero no-edit sanity check is exact ($\max |\text{rescue_gain}| = 0$). This is partial sufficiency rather than circuit recovery: the selected terminal features recover a measurable slice of the missing margin and beat both controls, but most of the U_IT,L_IT vs U_PT,L_IT gap remains.

Feature-rescue artifacts:

498 • results/exp43_feature_rescue_handoff/exp43_full_h100x8_clean_20260503_182947/analysis/exp43_rep
499 • results/exp43_feature_rescue_handoff/exp43_full_h100x8_clean_20260503_182947/analysis/primary_f
500 • results/exp43_feature_rescue_handoff/exp43_full_h100x8_clean_20260503_182947/analysis/exp43_fam

501 **Middle-to-terminal feature handoff.** This analysis tests whether upstream/preterminal computa-
502 tion drives the selected terminal features, rather than merely co-occurring with them. It uses the
503 three quality-gated feature families (Llama, Mistral, Qwen) and the same top-200 terminal causal
504 features. In the rescue direction, we start from the weak U_PT,L_IT hybrid and replace an upstream
505 MLP window with IT computation before running the IT terminal stack. In the degrade direction,
506 we start from native U_IT,L_IT and replace the same window with PT computation. The mediated
507 effect is the part of the margin change that disappears when the selected terminal features are ablated
508 in both the base and perturbed passes.

Handoff window / direction	Total margin effect	Terminal-feature-mediated part	Mediated fraction
mid-to-preterminal rescue into U_PT,L_IT	+1.714 [+1.634, +1.791]	+0.132 [+0.118, +0.147]	6.5%
mid-to-preterminal degradation of U_IT,L_IT	+3.570 [+3.427, +3.721]	+0.527 [+0.478, +0.576]	10.8%
terminal-entry upper-bound rescue	+5.147 [+4.936, +5.351]	+0.705 [+0.643, +0.767]	12.5%
terminal-entry upper-bound degradation	+5.147 [+4.953, +5.358]	+0.705 [+0.644, +0.762]	12.5%

509 The mid-to-preterminal mediated effect beats matched-random features (+0.188 rescue; +0.655
510 degradation), same-delta random directions (+0.115 rescue; +0.390 degradation), and top-active
511 noncausal features (+0.496 rescue; +0.928 degradation). The terminal-entry rows are upper-bound
512 sanity checks: they patch directly at the boundary into the terminal readout, so they should be larger
513 than nonterminal windows. The late-preterminal-only window is weaker, especially in Qwen's event-
514 permutation null, so the paper-facing claim is about mid-to-preterminal/preterminal handoff, not a
515 late-preterminal-only mechanism.

516 Handoff artifacts:

517 • results/exp44_middle_terminal_feature_handoff/exp44_primary_lmqa100_20260503_combined/analysis
518 • results/exp44_middle_terminal_feature_handoff/exp44_primary_lmqa100_20260503_combined/analysis
519 • results/exp44_middle_terminal_feature_handoff/exp44_primary_lmqa100_20260503_combined/analysis
520 • results/exp44_middle_terminal_feature_handoff/exp44_primary_lmqa100_20260503_combined/analysis

521 **Autointerp protocol.** Artifacts:

522 • results/exp39_causal_feature_interpretation/exp39_reinterp_specific_labels_ctrl_h100x8_20260503
523 • results/exp39_causal_feature_interpretation/exp39_reinterp_specific_labels_ctrl_h100x8_20260503
524 • results/exp39_causal_feature_interpretation/exp39_reinterp_specific_labels_ctrl_h100x8_20260503

525 Across 225 interpreted features from the clean terminal-crosscoder families, mean validation AU-
526 ROC is 0.886. We use these labels descriptively, not as causal evidence: the causal claim remains
527 the mediation, upstream-conditioning, and rescue results above. The paper-facing semantic check is
528 the narrower structure_readout bucket below, where a predeclared readable subset is edited and
529 tested against controls.

530 **Structure-readout bucket validation.** The structure-readout edit tests one readable subset from the
531 taxonomy rather than every label bucket. The predeclared structure_readout bucket contains 10
532 causal features across the three clean crosscoder families, with labels such as paragraph breaks, list
533 openings, answer boundaries, and field separators. This is not an N=10 statistical generalization
534 claim: the features are the predeclared edit set, while the test is whether the edited terminal readout
535 changes monotonically over prompts and families and beats matched controls. Editing this bucket
536 inside the same terminal crosscoder windows gives a monotone dose response in interaction drop;
537 matched-random and same-delta random controls are much smaller.

Edit strength alpha	Structure bucket	Matched random	Same-delta random
0.0	0.000	0.000	0.000
0.5	+0.039	-0.015	+0.001
1.0	+0.078	-0.028	+0.012
1.5	+0.125	-0.041	+0.020
2.0	+0.180	-0.048	+0.039

At alpha=2.0, per-family structure-bucket interaction drops are Llama +0.091, Mistral +0.339, and Qwen +0.110. Magnitudes are heterogeneous, but all three signs are positive. We use only this selective structure/readout result in the paper-facing feature-label validation. Other bucket edits were run as diagnostics but are not part of the evidence spine because their feature support is smaller or more domain-specific.

Structure-readout artifacts:

- results/exp41_causal_feature_bucket_steering/exp41_terminal_bucket_logit_full_h100x8_20260503_1
- results/exp41_causal_feature_bucket_steering/exp41_terminal_bucket_logit_full_h100x8_20260503_1
- results/exp41_causal_feature_bucket_steering/exp41_terminal_bucket_logit_full_h100x8_20260503_1

Appendix E: Late-Refinement Signatures

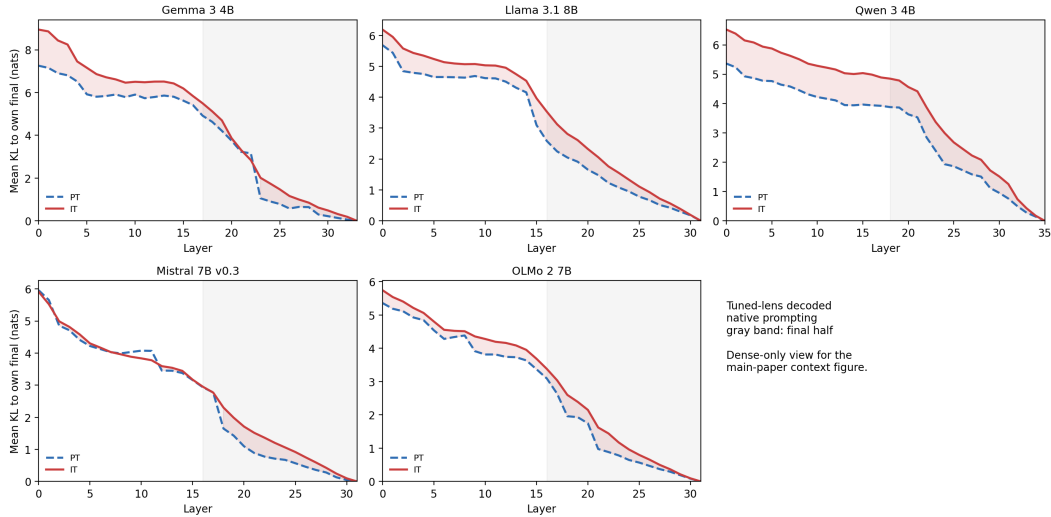
Layerwise stabilization. Primary artifacts:

- results/exp09_cross_model_observational_replication/data/exp9_summary.json
- results/exp09_cross_model_observational_replication/data/convergence_gap_values.json
- results/paper_synthesis/exp22_endpoint_deconfounded_table.csv

Endpoint-matched late KL estimates:

Metric	Estimate
Raw late 'KL(layer	
Tuned late 'KL(layer	
Remaining adjacent JS, IT - PT	+0.052 [+0.048, +0.057]
Future top-1 flips, IT - PT	+0.203 [+0.190, +0.215]

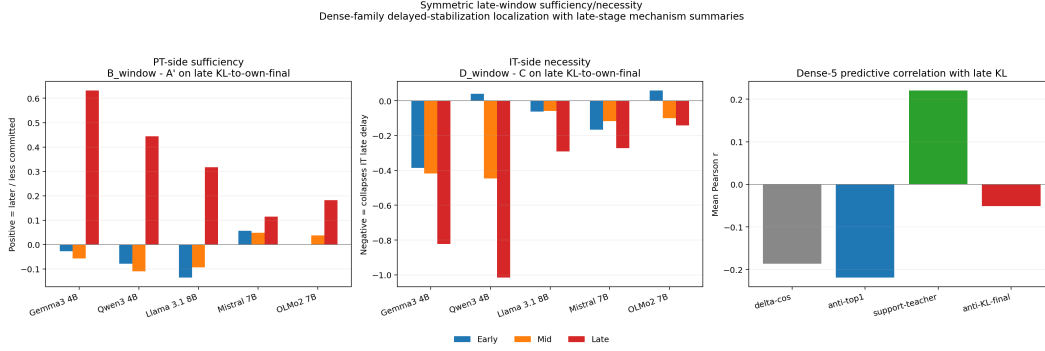
Layerwise Delayed Stabilization in Five Dense PT/IT Pairs



Late MLP localization. Primary artifacts:

- results/exp11_matched_prefix_mlp_graft/plots/exp11_exp3_600rand_v11_depthablation_full/depth_ab
- results/exp14_symmetric_matched_prefix_causality/exp13exp14_full_20260416/exp13_full_summary.js
- results/exp19_late_mlp_specificity_controls/exp19B_core120_h100x8_20260424_050421_analysis/exp1

The dense-family true late random-control comparison is +0.327 ([+0.298, +0.359]) for the learned late graft versus +0.003 ([−0.002, +0.008]) for matched random residual projections.



Appendix F: OLMo-2 Stage Case Study

Primary artifacts:

- results/exp35_olmo_base_anchored_stage_decomposition/exp35_full_olmo_stage_8a100_20260502_2300/
- results/exp35_olmo_base_anchored_stage_decomposition/exp35_full_olmo_stage_8a100_20260502_2300/
- results/exp35_olmo_base_anchored_stage_decomposition/exp35_full_olmo_stage_8a100_20260502_2300/
- results/exp35_olmo_base_anchored_stage_decomposition/exp35_full_olmo_stage_8a100_20260502_2300/

The primary stage analysis fixes the support to Base->RLVR first-divergence prefixes and scores every intermediate checkpoint against the same t_Base/t_RLVR contrast. This makes SFT, DPO, and RLVR cumulative estimates comparable on the same local support. The older adjacent-pair analysis is retained only as historical motivation because each adjacent contrast uses its own first-divergence support and token labels; those adjacent estimates are useful local contrasts, but they are not additive attributions to the final Base->RLVR contrast.

Stage on fixed Base->RLVR support	Upstream x late interaction	Relative to final contrast	Native top-1 picks t_RLVR
Base	0 by definition	0%	0.0%
SFT	+0.773 [+0.674, +0.873]	40.2% [37.6%, 42.8%]	61.0%
DPO	+1.629 [+1.473, +1.793]	84.7% [83.3%, 86.0%]	93.0%
RLVR/Instruct	+1.924 [+1.747, +2.104]	100%	99.7%

The fixed-support label-swap null passes the same orientation test as the main factorial: the observed RLVR interaction is +1.924, while the null 99.9th percentile is +0.382 ($p=5e-5$). Position ≥ 3 remains positive for all stages (+0.283, +0.677, +0.813). The result is a local lineage case study: in this released OLMo-2 path, the measured upstream-conditioned interaction is partly present in the SFT checkpoint, largely present in the DPO checkpoint, and strongest in the final RLVR/Instruct checkpoint.

Appendix G: Architecture and MoE Scope

Architecture heterogeneity and MoE. The Core-5 set covers dense transformer families only. It includes one Mistral family with an attention variant and one 32B Qwen scale check, but it is not a controlled architecture sweep. MoE generalization remains open: DeepSeek-V2-Lite is artifact-only because expert routing, expert swaps, and sparse activation patterns require additional controls beyond the dense-stack factorial used here. The clean follow-up is to run the same first-divergence factorial across multiple MoE base/instruct pairs with expert-routing controls and to separate attention-pattern effects from post-training recipe effects.

Appendix H: Reproducibility and Artifact Map

Claim	Command/script family
Core-5 upstream x late interaction	scripts/analysis/build_exp23_core5_synthesis.py; Core-5 figure generated from exp23_core5_family_effects.
Position sensitivity	same plus scripts/analysis/analyze_first_divergence_position_sensitivity.py

Claim	Command/script family
Label-swap null	scripts/analysis/analyze_exp23_compatibility_permutation.py
Off-manifold sanity audit	scripts/analysis/analyze_exp23_offmanifold_sanity.py
Hybrid-state validation	scripts/run/run_exp36_offmanifold_validation_runpod.sh; scripts/analysis/analyze_exp36_offmanifold_v
Selection baselines and token-support control	scripts/run/run_exp37_random_prefix_baseline_runpod.sh; scripts/analysis/analyze_exp37_random_prefix
Selected-support audit	scripts/analysis/analyze_first_divergence_token_support.py
Pre-late commitment control	scripts/analysis/analyze_exp40_prelate_commitment_control.py; exact collector in src/poc/exp40_prelate_co
Behavioral bridge from margin to output	src/poc/exp45_behavioral_bridge/; scripts/analysis/analyze_exp45_behavioral_bridge.py; scripts/scorin
Endpoint-matched convergence gap	scripts/analysis/build_exp22_endpoint_deconfounded_synthesis.py
Late MLP random control	late-random-control analysis scripts
Identity/margin handoff	scripts/analysis/build_exp20_exp21_handoff_synthesis.py
Terminal-depth audit	scripts/analysis/analyze_exp31_terminal_depth_factorial.py
Middle-to-terminal feature handoff	scripts/run/run_exp44_middle_terminal_feature_handoff_runpod.sh; src/poc/exp44_middle_terminal_featu
Terminal MLP audit	scripts/analysis/analyze_exp33_terminal_identity_margin.py
Terminal crosscoder mediation	scripts/analysis/analyze_exp34_dense5_final_readout_crosscoder.py; crosscoder hardening analysis
Terminal feature upstream-conditioning	src/poc/exp42_terminal_feature_upstream_conditioning/; scripts/analysis/analyze_exp42_terminal_featu
Terminal feature rescue	src/poc/exp43_feature_rescue_handoff/; scripts/plot/plot_exp43_feature_rescue_handoff.py
Feature autointerp and taxonomy	src/poc/exp39_causal_feature_interpretation/; scripts/analysis/exp39_causal_paper_taxonomy_llm.py
Structure-readout bucket validation	src/poc/exp41_causal_feature_bucket_steering/; structure-readout analysis outputs
OLMo fixed-support stage case study	scripts/analysis/analyze_exp35_olmo_base_anchored_stage_decomposition.py; scripts/analysis/build_exp

588 All full reruns use bf16 inference and deterministic greedy decoding unless a script states other-
589 wise. The summary audit is CPU-only and reads committed JSON/CSV artifacts. Reproducing raw
590 4B-8B intervention records requires multiple 80GB A100/H100 jobs; reproducing Qwen2.5 32B
591 additionally requires the multi-GPU run or the committed paper-facing synthesis artifacts.

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